

Rahul Bajaj

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WORK EXPERIENCE

Philips

Aug. 2023 – Present

Software Development Engineer

Bangalore, India

- **Frontend Platform:** Directed a 4-member team to architect an enterprise launchpad using **React, TypeScript,** and **AWS Serverless**, increasing deployment frequency by **35%** for **10k+** daily users.
- **MFE Architecture:** Engineered MFE solutions using **Module Federation** and **Vite**, slashing session-related errors by **40%** while improving page load speeds by **1.2s**.
- **UI Design Systems:** Designed dynamic forms and workflow-based UI components; improved component reusability by **60%**, saving **150+** engineering hours per quarter.
- **Cloud Infrastructure:** Managed **Java Spring Boot** microservices on **AWS EKS** for IoT medical device connectivity, maintaining **99.9%** uptime and supporting **50k+** concurrent connections.
- **System Architecture:** Implemented **Domain-Driven Design (DDD)** and **Design Patterns (Strategy, Proxy, Facade)**, resulting in a **25%** reduction in technical debt.

Philips

Feb. 2023 – July 2023

SDE Intern

Bangalore, India

- **Cloud Automation:** Built a secure **AWS Serverless** application using **Python (Boto3)**, **API Gateway**, and **PostgreSQL/DynamoDB**, reducing processing time from **3 days** to **15 minutes**.
- **Performance Tuning:** Optimized **PostgreSQL** queries and indexing, improving retrieval performance by **55%** and eliminating timeouts during peak hours.
- **Agile Collaboration:** Participated in an **Agile SDLC (Scrum)** environment, delivering **100%** of sprint commitments while refining technical requirements.

Projects

Spring Code Forger

- **Agentic Codegen:** Built a multi-agent system that ingests **Swagger/OpenAPI** specs and auto-generates production-ready **Spring Boot** CRUD scaffolding (controllers, services, repos, DTOs).
- **LLM Orchestration:** Designed an agent-per-layer pipeline using **LangChain + Claude**, cutting boilerplate authoring time by **~70%** for new microservices.

TestForge

- **Test Automation:** Built a multi-agent workflow that generates **Selenium/Playwright** automation tests from plain-English user stories, reducing manual QA effort.
- **Agentic Pipeline:** Coordinated planner → writer → validator agents for self-healing test generation against live app DOM.

EDUCATION

Chandigarh Group of Colleges | B,tech in Computer Science & Engineering | July, 2023

- **Leadership:** Captain of the University Badminton Team (led to multiple tournament wins).
- **Management:** Finance head for flagship festival (**500+** delegates); Lead Anchor for university events.

CERTIFICATIONS, SKILLS & INTERESTS

- **Languages & Backend:** TypeScript, Java, Python, SQL, Spring Boot, REST APIs, Microservices.
- **Frontend & Architecture:** React.js, Micro Frontends (MFE), Module Federation, Vite, Next.js.
- **Cloud & DevOps:** AWS (Lambda, S3, DynamoDB, RDS), EKS, CI/CD, Infrastructure as Code (IaC).
- **Testing & Design:** Design Patterns, Domain-Driven Design (DDD), Karate, Jest, Unit Testing.
- **Interests:** Badminton (University Team), Weightlifting, Adventure Sports, Reading.